

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ELECTRICAL AND ELECTRONIC ENGINEERING



WIRELESS COMMUNICATION

PAPR–Performance Tradeoff: A Comparative Study of OFDM and SC-FDMA

NGUYỄN ĐỖ HOÀNG MINH

minh.ndh252697M@sis.hust.edu.vn

Program: Master of Science in Electronics Engineering

Supervisor: Prof. Nguyễn Văn Đức _____

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LIST OF ABBREVIATIONS

AWGN	Additive White Gaussian Noise
BER	Bit Error Rate
B5G	Beyond Fifth Generation
CCDF	Complementary Cumulative Distribution Function
CP	Cyclic Prefix
DAC	Digital-to-Analog Converter
DFT	Discrete Fourier Transform
FDMA	Frequency Division Multiple Access
FFT	Fast Fourier Transform
IDFT	Inverse Discrete Fourier Transform
IFFT	Inverse Fast Fourier Transform
ISI	Inter-Symbol Interference
LTE	Long Term Evolution
LP-OFDMA	Linearly Precoded Orthogonal Frequency Division Multiple Access
MIMO	Multiple-Input Multiple-Output
NR	New Radio
OFDM	Orthogonal Frequency Division Multiplexing
PAPR	Peak-to-Average Power Ratio
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RF	Radio Frequency
SC-FDMA	Single-Carrier Frequency Division Multiple Access
SC-OFDM	Single-Carrier Orthogonal Frequency Division Multiplexing
SNR	Signal-to-Noise Ratio
THz	Terahertz
ZF	Zero-Forcing

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ABSTRACT

This report presents a comparative study of Orthogonal Frequency Division Multiplexing (OFDM) and Single-Carrier Frequency Division Multiple Access (SC-FDMA), two important multicarrier transmission techniques widely used in modern wireless communication systems. OFDM is well known for its high spectral efficiency, robustness against multipath fading, and efficient frequency-domain equalization. However, one major limitation of OFDM is its high Peak-to-Average Power Ratio (PAPR), which reduces power amplifier efficiency and introduces nonlinear distortion in practical transmission systems.

SC-FDMA was introduced to address the high-PAPR problem of OFDM by applying a Discrete Fourier Transform (DFT) spreading stage prior to OFDM modulation. This additional precoding operation allows SC-FDMA to retain many advantages of OFDM while exhibiting single-carrier-like transmission characteristics and lower PAPR. Consequently, SC-FDMA is more suitable for uplink communication in battery-powered mobile devices, where transmitter power efficiency is critical.

In this study, the operating principles of OFDM and SC-FDMA are analyzed theoretically and evaluated through MATLAB-based simulations. The systems are implemented using 16-QAM modulation over multipath Rayleigh fading channels with Zero Forcing (ZF) equalization. Performance evaluation is conducted using Bit Error Rate (BER) and Complementary Cumulative Distribution Function (CCDF) analysis of PAPR. Simulation results show that both systems achieve comparable BER performance under identical channel conditions, while SC-FDMA demonstrates significantly lower PAPR than OFDM.

The results highlight the tradeoff between transmission performance and power efficiency, and explain the practical adoption of OFDM in downlink systems and SC-FDMA in LTE uplink communications.

CHAPTER 1. INTRODUCTION

1.1 Background

The rapid development of modern wireless communication systems has created a growing demand for high data rate transmission, efficient spectrum utilization, low latency, and reliable communication under challenging channel conditions. Applications such as mobile broadband, video streaming, Internet of Things (IoT), and fifth-generation (5G) wireless networks require transmission techniques capable of operating efficiently in multipath fading environments while maintaining acceptable complexity and power consumption.

In wireless channels, transmitted signals often experience reflection, diffraction, and scattering due to surrounding objects such as buildings, vehicles, and terrain. As a result, multiple delayed copies of the signal arrive at the receiver simultaneously, producing a phenomenon known as multipath fading. This effect causes inter-symbol interference (ISI), which significantly degrades communication performance.

$$r(t) = s(t) * h(t) + n(t) \quad (1.1)$$

where $s(t)$ is the transmitted signal, $h(t)$ is the wireless channel impulse response, and $n(t)$ represents additive noise.

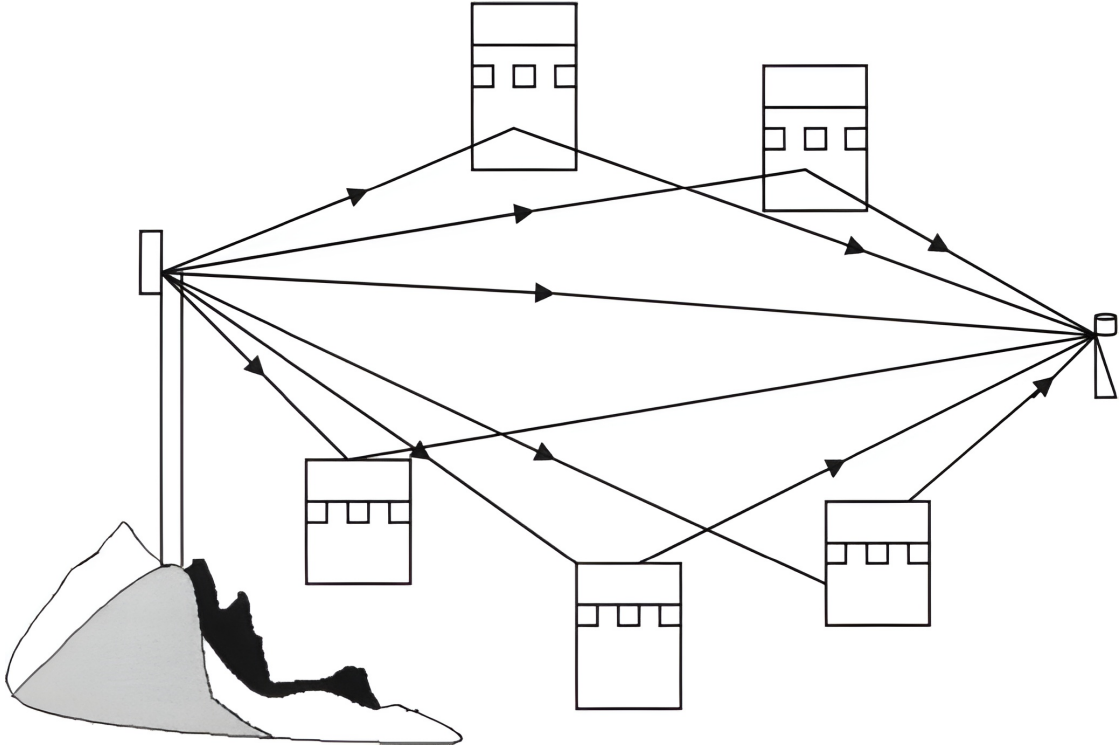


Figure 1.1 Illustration of multipath propagation phenomena

Orthogonal Frequency Division Multiplexing (OFDM) was developed to combat frequency-selective fading by dividing a high-rate data stream into many low-rate parallel subcarriers. Since each subcarrier experiences a nearly flat fading channel, equalization becomes significantly simpler compared to conventional single-carrier systems.

The OFDM transmitted signal can be represented mathematically using the Inverse Fast Fourier Transform (IFFT):

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X_k e^{j\frac{2\pi kn}{N}} \quad (1.2)$$

where:

- X_k denotes the modulated data symbol on the k -th subcarrier,
- N is the FFT size,
- $x[n]$ is the time-domain OFDM signal.

The orthogonality between subcarriers allows efficient spectrum usage while avoiding inter-carrier interference (ICI).

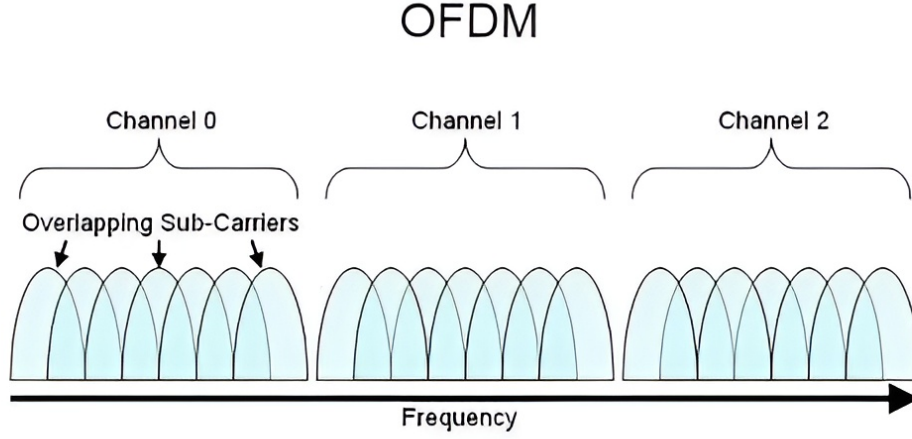


Figure 1.2 Orthogonal overlapping subcarriers in OFDM systems

Although OFDM provides excellent spectral efficiency and robustness against multipath fading, one major drawback is its high Peak-to-Average Power Ratio (PAPR). Since an OFDM signal is formed by the superposition of many independently modulated subcarriers, constructive interference occasionally produces extremely large signal peaks.

The PAPR of a transmitted signal is defined as:

$$\text{PAPR} = \frac{\max |x(t)|^2}{E\{|x(t)|^2\}} \quad (1.3)$$

where:

- $\max |x(t)|^2$ is the peak instantaneous power,
- $E\{|x(t)|^2\}$ denotes the average signal power.

High PAPR reduces the efficiency of power amplifiers because the amplifier must operate with large power back-off to avoid nonlinear distortion and signal clipping.

To address this limitation, Single-Carrier Frequency Division Multiple Access (SC-FDMA) was introduced as an uplink transmission technique for Long Term Evolution (LTE) systems. SC-FDMA combines the advantages of single-carrier transmission and OFDM by inserting a Discrete Fourier Transform (DFT) spreading operation before OFDM modulation.

The DFT spreading operation is expressed as:

$$X[m] = \sum_{n=0}^{M-1} d[n] e^{-j \frac{2\pi mn}{M}} \quad (1.4)$$

where:

- $d[n]$ represents the input modulation symbols,
- $X[m]$ denotes the spread frequency-domain symbols,
- M is the number of occupied subcarriers.

By spreading information symbols across multiple subcarriers, SC-FDMA produces a signal with lower envelope fluctuation and lower PAPR compared to conventional OFDM.

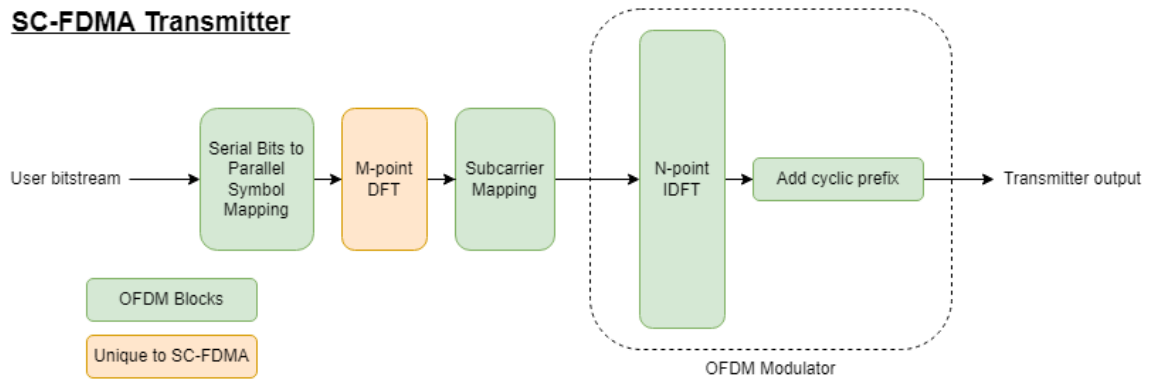


Figure 1.3 Comparison between OFDM and SC-FDMA transmitter structures

For this reason, SC-FDMA was selected for LTE uplink transmission, where mobile devices require high power efficiency and long battery lifetime.

1.2 Problem Statement

Despite its many advantages, OFDM suffers from high PAPR, which creates practical implementation challenges in wireless transmitters. Large signal peaks reduce power amplifier efficiency and increase nonlinear distortion, leading to spectral regrowth and degraded transmission quality.

Although SC-FDMA was developed to reduce the PAPR problem, its performance characteristics differ from OFDM due to the DFT spreading operation. Therefore, a detailed comparison between OFDM and SC-FDMA is necessary to understand the tradeoff between transmission performance, power efficiency, and implementation complexity.

Furthermore, multipath fading channels introduce frequency-selective distortion that affects both systems differently. Equalization techniques must therefore be applied at the receiver to recover transmitted data and maintain reliable communication.

1.3 Objectives

The main objectives of this report are summarized as follows:

- To study the operating principles of OFDM and SC-FDMA systems.
- To analyze the impact of multipath fading channels on system performance.
- To implement OFDM and SC-FDMA transmission systems using MATLAB.
- To evaluate system performance using Bit Error Rate (BER) analysis under different Signal-to-Noise Ratio (SNR) conditions.
- To compare the Peak-to-Average Power Ratio (PAPR) performance using Complementary Cumulative Distribution Function (CCDF) analysis.
- To discuss the engineering tradeoff between transmission reliability and power efficiency.

1.4 Scope of the Study

This report focuses on the simulation and analysis of OFDM and localized SC-FDMA systems under multipath Rayleigh fading channels. The systems are implemented using 16-QAM modulation with identical FFT size, cyclic prefix length, active subcarrier allocation, and transmission bandwidth for fair comparison.

The receiver employs Zero Forcing (ZF) equalization to compensate for channel distortion in the frequency domain.

The study mainly focuses on:

- BER performance comparison,
- PAPR analysis,
- multipath fading effects,
- frequency-domain equalization.

The following topics are outside the scope of this report:

- channel coding,
- MIMO transmission,
- adaptive modulation,
- interleaved SC-FDMA mapping,
- advanced equalization techniques such as MMSE.

1.5 Report Structure

This report is organized into the following chapters:

- Chapter 1 introduces the background, motivation, objectives, and scope of the study.
- Chapter 2 presents the theoretical foundations of OFDM, SC-FDMA, multipath fading, PAPR, and equalization techniques.
- Chapter 3 describes the simulation methodology, system parameters, channel model, and MATLAB implementation.
- Chapter 4 presents simulation results and discusses BER and PAPR performance comparisons between OFDM and SC-FDMA.
- Chapter 5 summarizes the main conclusions and proposes future research directions.

CHAPTER 2. THEORETICAL BACKGROUND

2.1 Overview of Multicarrier Transmission

Modern wireless communication systems operate in environments affected by multipath propagation, delay spread, Doppler effects, and additive noise. These impairments introduce inter-symbol interference (ISI) and degrade transmission reliability, especially at high data rates.

Conventional single-carrier systems require complex time-domain equalization to compensate for frequency-selective fading channels. As bandwidth increases, equalizer complexity grows significantly.

To overcome this limitation, multicarrier transmission techniques divide the available bandwidth into multiple narrowband orthogonal subcarriers. Instead of transmitting one high-rate stream, the input data stream is split into several parallel low-rate streams.

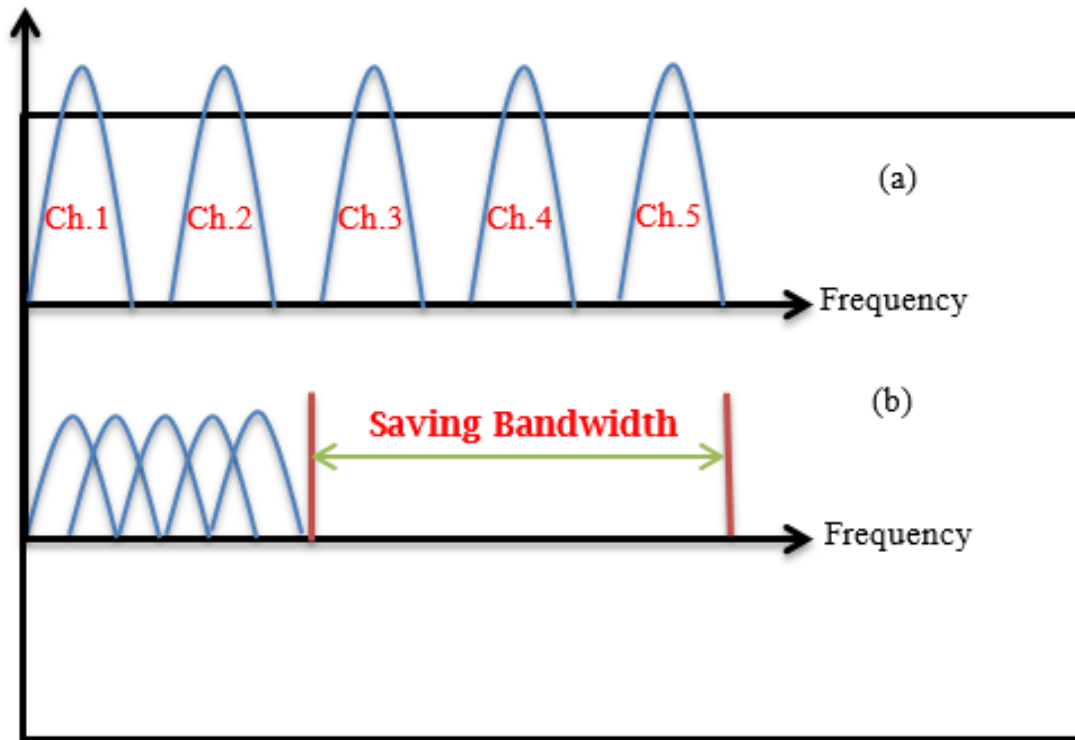


Figure 2.1 Multicarrier technique versus Orthogonal Multicarrier technique

If the total system bandwidth is B and the number of subcarriers is N , then the bandwidth of each subcarrier becomes:

$$\Delta f = \frac{B}{N} \quad (2.1)$$

As N increases, each subcarrier experiences nearly flat fading, which greatly simplifies equalization.

Orthogonal Frequency Division Multiplexing (OFDM) is the most widely used multicarrier transmission scheme and forms the basis of many modern wireless standards such as Wi-Fi, LTE, DVB, and 5G systems.

2.2 Wireless Multipath Fading Channels

Wireless signals propagate through multiple paths due to reflection, diffraction, and scattering from surrounding objects. The received signal therefore becomes the superposition of several delayed and attenuated copies of the transmitted waveform.

The channel impulse response of a multipath fading channel can be modeled as:

$$h(t) = \sum_{l=0}^{L-1} \alpha_l \delta(t - \tau_l) \quad (2.2)$$

where:

- L is the number of channel paths,
- α_l represents the complex gain of the l -th path,
- τ_l is the propagation delay,
- $\delta(\cdot)$ is the Dirac delta function.

The received signal is therefore expressed as:

$$r(t) = s(t) * h(t) + n(t) \quad (2.3)$$

where:

- $s(t)$ is the transmitted signal,
- $h(t)$ is the channel response,
- $n(t)$ denotes additive white Gaussian noise (AWGN).

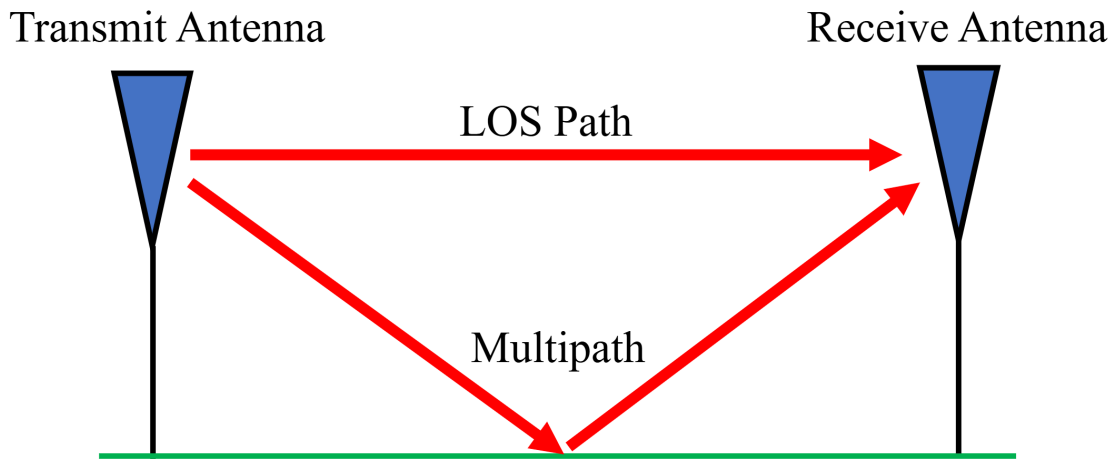


Figure 2.2 Multipath fading channel model

When no dominant line-of-sight component exists, the channel envelope typically follows a Rayleigh distribution. Such channels are commonly called Rayleigh fading channels.

Frequency-selective fading occurs when different frequency components of the signal experience different channel gains. OFDM and SC-FDMA were developed specifically to combat this problem efficiently.

2.3 Orthogonal Frequency Division Multiplexing (OFDM)

Orthogonal Frequency Division Multiplexing (OFDM) is a multicarrier transmission technique in which a high-rate data stream is divided into multiple lower-rate parallel streams and transmitted simultaneously over many orthogonal subcarriers.

Instead of using one high-bandwidth carrier, OFDM partitions the available spectrum into narrowband subchannels. Because each subcarrier occupies a small bandwidth, the effect of frequency-selective fading is significantly reduced, and each subcarrier can be approximated as experiencing flat fading.

OFDM has become one of the most important waveform technologies in modern wireless communication systems and has been adopted in standards such as:

- Wi-Fi (IEEE 802.11)
- LTE Downlink
- 5G NR
- DVB broadcasting systems
- ADSL and broadband wired communications

The basic OFDM transmitter structure is shown in Figure 2.3.

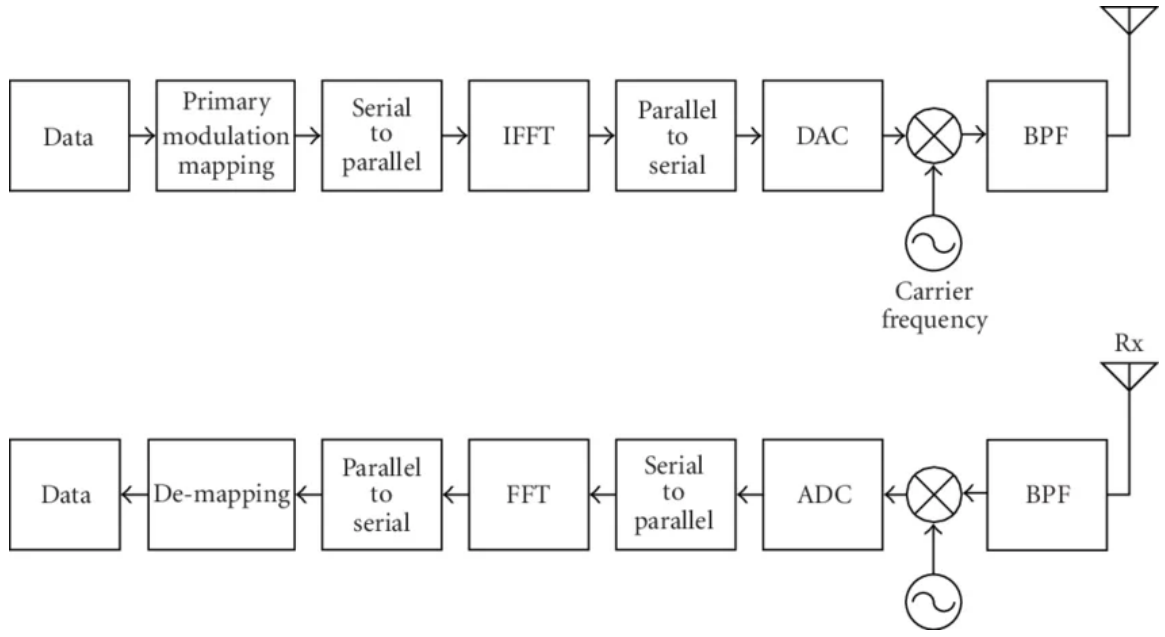


Figure 2.3 OFDM transmitter-receiver structure

At the transmitter, the incoming serial bit stream is first converted into parallel groups of bits and mapped into complex modulation symbols using modulation schemes such as QPSK, 16-QAM, or 64-QAM.

Assume that:

$$X = [X[0], X[1], \dots, X[N-1]] \quad (2.4)$$

represents the frequency-domain symbols assigned to the OFDM subcarriers.

The OFDM modulator converts these symbols into a time-domain waveform using the Inverse Fast Fourier Transform (IFFT):

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi kn}{N}} \quad (2.5)$$

where:

- $X[k]$ is the complex symbol transmitted on subcarrier k ,
- N is the total number of subcarriers,
- $x[n]$ is the discrete-time OFDM signal.

Equation 2.5 shows that the OFDM waveform is formed by the superposition of many orthogonal sinusoidal carriers.

Because the subcarriers are orthogonal, they can overlap in the frequency domain without causing inter-carrier interference (ICI). This allows OFDM to achieve very high spectral efficiency compared to conventional single-carrier systems.

At the receiver, the Fast Fourier Transform (FFT) operation converts the received signal back into the frequency domain:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi kn}{N}} \quad (2.6)$$

The FFT efficiently separates the orthogonal subcarriers and recovers the transmitted data symbols.

The major advantages of OFDM include:

- High spectral efficiency
- Excellent robustness against multipath fading
- Simple frequency-domain equalization
- Efficient implementation using FFT/IFFT algorithms

However, OFDM also suffers from several practical limitations:

- High Peak-to-Average Power Ratio (PAPR)
- Sensitivity to carrier frequency offset
- Sensitivity to synchronization errors

Among these limitations, high PAPR is considered one of the most critical issues in practical transmitter implementation.

2.4 Orthogonality of Subcarriers

The key principle behind OFDM is orthogonality between subcarriers.

Although the subcarriers overlap in the frequency domain, they remain mathematically independent over one symbol duration. This property enables multiple subcarriers to share the same bandwidth efficiently without mutual interference.

Two continuous-time subcarriers are orthogonal if:

$$\int_0^T e^{j2\pi(f_k - f_m)t} dt = 0 \quad k \neq m \quad (2.7)$$

where:

- f_k and f_m are different subcarrier frequencies,
- T is the OFDM symbol duration.

To satisfy this condition, adjacent subcarriers are separated by:

$$\Delta f = \frac{1}{T} \quad (2.8)$$

This spacing guarantees that the peak of one subcarrier coincides with the zero crossings of neighboring subcarriers.

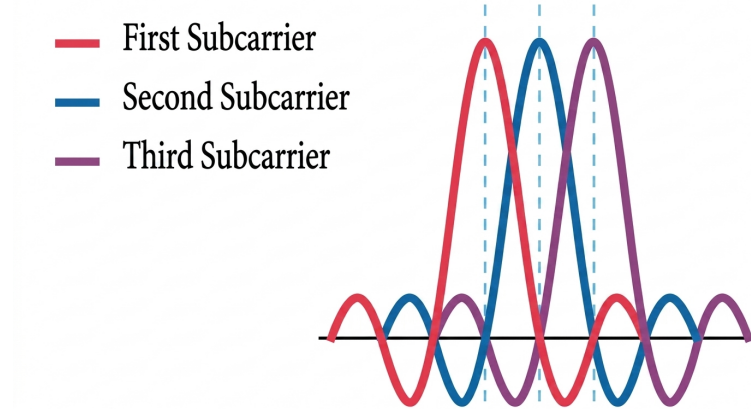


Figure 2.4 Orthogonality between adjacent OFDM subcarriers

Figure 2.4 illustrates how orthogonal subcarriers overlap spectrally while remaining separable at the receiver.

This property significantly improves spectrum utilization compared to traditional Frequency Division Multiplexing (FDM), where large guard bands are required between channels.

2.5 Cyclic Prefix (CP)

Wireless channels often contain multiple delayed signal paths caused by reflection, diffraction, and scattering.

As a result, delayed replicas of previous OFDM symbols may overlap with the current symbol and create Inter-Symbol Interference (ISI).

To combat ISI, OFDM inserts a Cyclic Prefix (CP) before each OFDM symbol.

The CP is created by copying the final portion of the OFDM symbol and appending it to the beginning of the transmitted block.

If the CP length exceeds the maximum channel delay spread, the linear convolution between the channel and transmitted signal becomes circular convolution:

$$y[n] = x[n] \circledast h[n] \quad (2.9)$$

where:

- $x[n]$ is the transmitted OFDM symbol,
- $h[n]$ is the channel impulse response,
- $y[n]$ is the received signal.

This transformation is extremely important because circular convolution in the time domain corresponds to simple multiplication in the frequency domain:

$$Y[k] = X[k]H[k] \quad (2.10)$$

As a result, equalization becomes very simple since each subcarrier only requires a one-tap equalizer:

$$\hat{X}[k] = \frac{Y[k]}{H[k]} \quad (2.11)$$

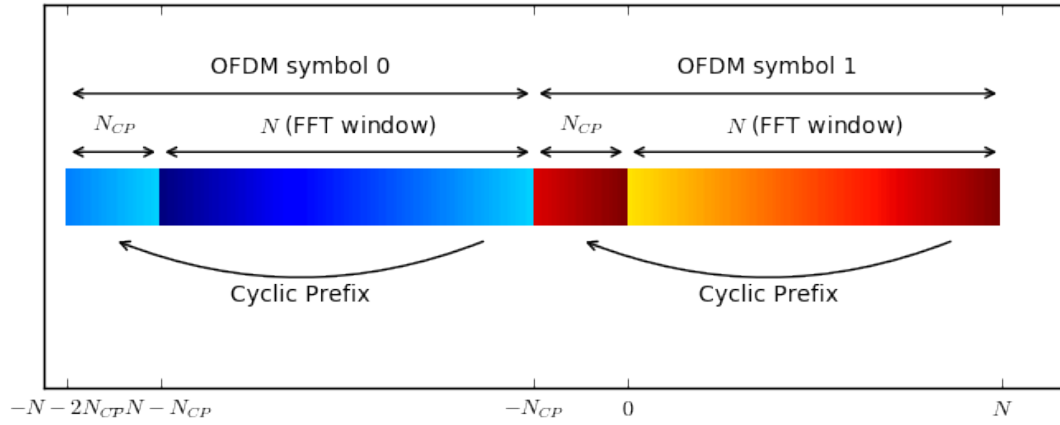


Figure 2.5 Cyclic prefix insertion in OFDM systems

The figure above illustrates two consecutive OFDM symbols, where each symbol contains its own cyclic prefix. The CP is a replica of the final samples of the OFDM block. At the receiver, the FFT window begins after the cyclic prefix is removed. If the CP duration is sufficiently long, delayed channel components remain inside the guard interval and do not corrupt the useful OFDM symbol.

Although the cyclic prefix improves robustness against multipath fading, it also introduces overhead and slightly reduces spectral efficiency.

2.6 Single-Carrier Frequency Division Multiple Access (SC-FDMA)

Single-Carrier Frequency Division Multiple Access (SC-FDMA) is a modified version of OFDM designed to reduce Peak-to-Average Power Ratio (PAPR).

SC-FDMA is also commonly referred to as:

- DFT-spread OFDM (DFT-s-OFDM)
- Single-Carrier OFDM (SC-OFDM)

Unlike conventional OFDM, SC-FDMA inserts a Discrete Fourier Transform (DFT) operation before subcarrier mapping.

This additional processing spreads each modulation symbol across multiple subcarriers before OFDM modulation.

Because of this spreading effect, the transmitted waveform exhibits single-carrier characteristics with reduced envelope fluctuation.

The transmitter and receiver structures are illustrated in Figure 2.6.

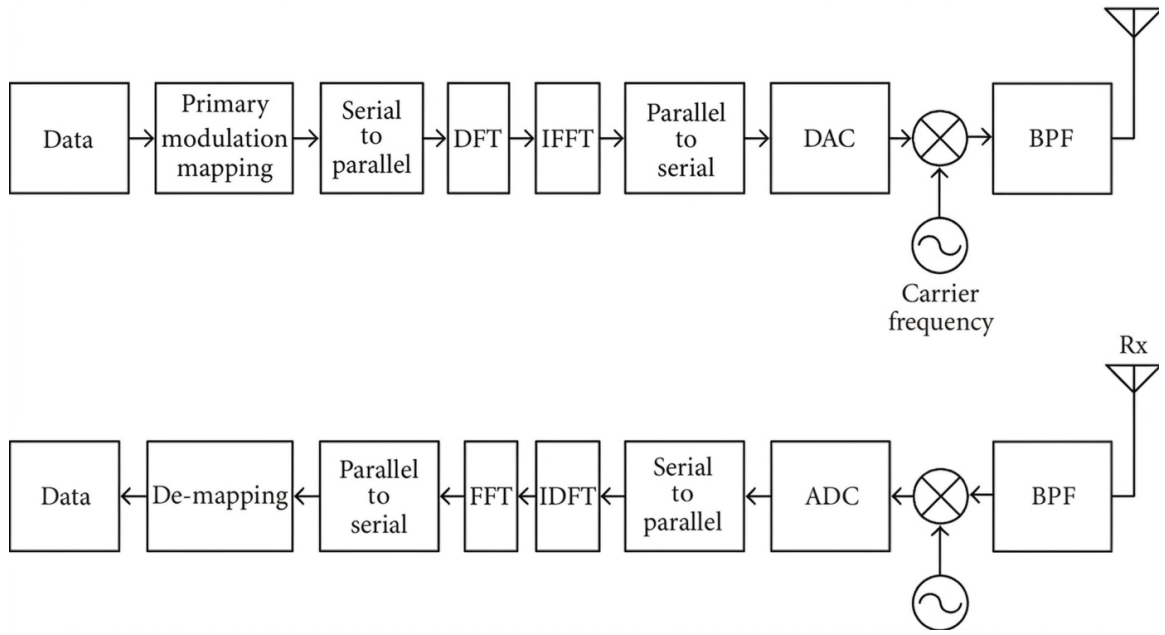


Figure 2.6 SC-FDMA transmitter-receiver structure

At the transmitter, the modulated symbols are first transformed using an M -point DFT:

$$X[m] = \sum_{n=0}^{M-1} d[n] e^{-j \frac{2\pi mn}{M}} \quad (2.12)$$

where:

- $d[n]$ is the input modulation symbol,
- $X[m]$ is the DFT-spread symbol,
- M is the number of occupied subcarriers.

The spread symbols are then mapped onto OFDM subcarriers before the IFFT operation.

The resulting time-domain signal becomes:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j \frac{2\pi kn}{N}} \quad (2.13)$$

Although Equation 2.13 appears mathematically similar to OFDM, the important difference is that the input symbols have already been precoded by the DFT operation.

This DFT spreading distributes information symbols across the occupied subcarriers and significantly reduces instantaneous signal peaks.

At the receiver, the cyclic prefix is removed and FFT processing is performed similarly to OFDM.

After subcarrier demapping, an inverse DFT (IDFT) operation reconstructs the original modulation symbols:

$$d[n] = \frac{1}{M} \sum_{m=0}^{M-1} X[m] e^{j \frac{2\pi mn}{M}} \quad (2.14)$$

The complete receiver chain can therefore be summarized as:

Remove CP → FFT → Subcarrier Demapping → IDFT → Symbol Detection

One major advantage of SC-FDMA is lower PAPR compared to OFDM.

In OFDM, multiple independently phased subcarriers may occasionally add constructively, creating very large peaks in the transmitted waveform.

In SC-FDMA, DFT spreading introduces correlation between subcarriers, making the transmitted waveform resemble a single-carrier signal with smaller amplitude variation.

This property improves power amplifier efficiency and reduces nonlinear distortion.

For this reason, SC-FDMA was selected as the uplink waveform in LTE mobile communication systems, where battery efficiency is critically important.

SC-FDMA also exhibits improved robustness against localized spectral nulls caused by frequency-selective fading.

In OFDM, severe fading on one subcarrier directly destroys the information carried on that subcarrier.

In SC-FDMA, information symbols are spread across multiple subcarriers. Therefore, fading on a single subcarrier affects all symbols only partially rather than completely destroying one symbol.

However, SC-FDMA also introduces several tradeoffs:

- Additional DFT/IDFT processing complexity
- More complicated subcarrier mapping structures
- Slightly less flexibility compared to OFDM

Despite these tradeoffs, the lower PAPR characteristic makes SC-FDMA highly suitable for uplink mobile transmission systems.

2.7 Localized SC-FDMA Mapping

There are two main SC-FDMA subcarrier mapping methods:

- Localized FDMA (LFDMA)
- Interleaved FDMA (IFDMA)

In localized mapping, consecutive subcarriers are allocated to a user:

$$k = 0, 1, 2, \dots, M - 1 \quad (2.15)$$

In interleaved mapping, subcarriers are distributed uniformly across the entire bandwidth:

$$k = 0, L, 2L, \dots, (M - 1)L \quad (2.16)$$

where L is the subcarrier spacing interval.

This report focuses only on localized SC-FDMA because it is more commonly used in LTE uplink systems.

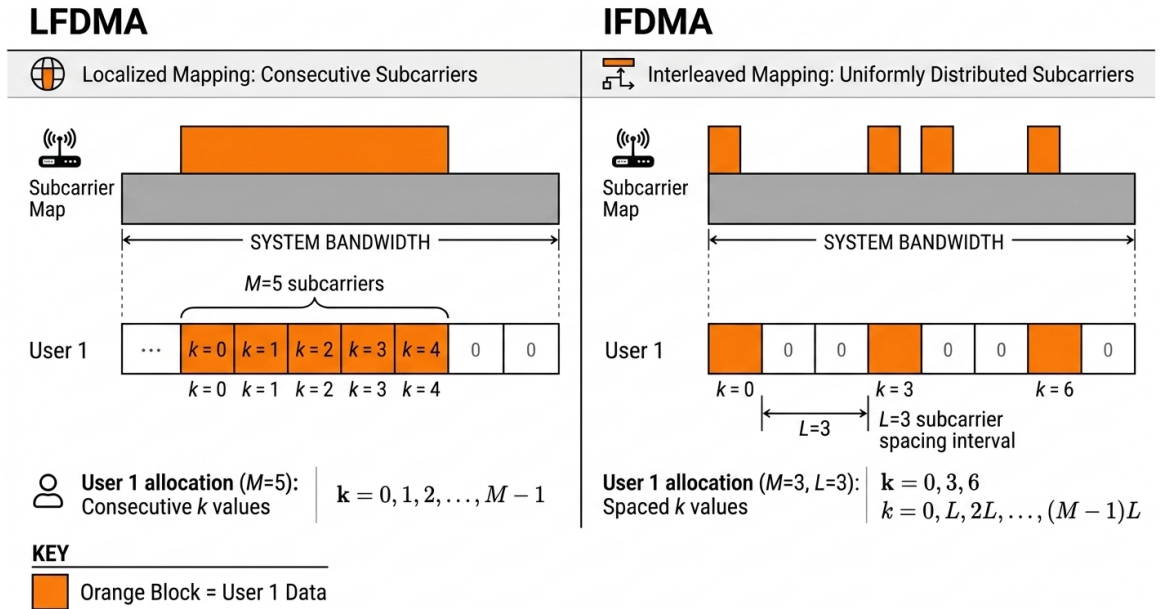


Figure 2.7 Localized and interleaved SC-FDMA subcarrier allocation

2.8 Peak-to-Average Power Ratio (PAPR)

PAPR measures the ratio between the maximum instantaneous signal power and its average power.

Mathematically:

$$\text{PAPR} = \frac{\max |x[n]|^2}{E\{|x[n]|^2\}} \quad (2.17)$$

PAPR is usually expressed in decibels:

$$\text{PAPR}_{dB} = 10 \log_{10} \left(\frac{\max |x[n]|^2}{E\{|x[n]|^2\}} \right) \quad (2.18)$$

In OFDM systems, independently phased subcarriers occasionally combine constructively, creating large signal peaks.

According to the central limit theorem, the OFDM signal amplitude tends toward a Gaussian distribution as the number of subcarriers increases.

SC-FDMA reduces this effect because DFT spreading introduces correlation between transmitted samples, producing a waveform closer to conventional single-carrier transmission.

2.9 Complementary Cumulative Distribution Function (CCDF)

The Complementary Cumulative Distribution Function (CCDF) is commonly used to evaluate PAPR statistics.

CCDF represents the probability that the signal PAPR exceeds a threshold value P_0 :

$$\text{CCDF}(P_0) = \Pr(\text{PAPR} > P_0) \quad (2.19)$$

A lower CCDF curve indicates lower probability of high signal peaks and better power amplifier efficiency.

2.10 Frequency-Domain Equalization

Multipath fading distorts the transmitted signal differently across subcarriers. Equalization is therefore necessary to compensate for channel effects.

After FFT demodulation, the received signal on the k -th subcarrier is:

$$Y[k] = H[k]X[k] + N[k] \quad (2.20)$$

where:

- $H[k]$ is the channel frequency response,

- $X[k]$ is the transmitted symbol,
- $N[k]$ is additive noise.

In Zero Forcing (ZF) equalization, the estimated transmitted symbol is:

$$\hat{X}[k] = \frac{Y[k]}{H[k]} \quad (2.21)$$

ZF equalization completely removes channel distortion under ideal channel estimation conditions. However, it may significantly amplify noise when $H[k]$ becomes very small.

2.11 Bit Error Rate (BER)

Bit Error Rate (BER) measures communication reliability by comparing the number of incorrectly detected bits to the total transmitted bits.

$$\text{BER} = \frac{N_e}{N_t} \quad (2.22)$$

where:

- N_e is the number of erroneous bits,
- N_t is the total number of transmitted bits.

Lower BER indicates better system performance.

In this report, BER under different Signal-to-Noise Ratio (SNR) conditions is used to compare OFDM and SC-FDMA performance in multipath fading channels.

2.12 Comparison Summary

Table 2.1 Comparison between OFDM and SC-FDMA

Criterion	OFDM	SC-FDMA
Transmission Structure	Pure multicarrier	DFT-spread multicarrier
PAPR	High	Lower
Power Amplifier Efficiency	Lower	Higher
Equalization Complexity	Simple	Simple
Frequency Diversity	Limited to subcarrier	Spread across subcarriers
Implementation Complexity	Lower	Slightly Higher
LTE Usage	Downlink	Uplink

2.13 Chapter Summary

This chapter presented the theoretical foundations of OFDM and SC-FDMA, including multicarrier transmission principles, multipath fading channels, orthogonality, cyclic prefix operation, DFT spreading, PAPR analysis, CCDF evaluation, and frequency-domain equalization.

The next chapter presents the MATLAB simulation model and analyzes the BER and PAPR performance of OFDM and SC-FDMA systems under multipath Rayleigh fading channels.

CHAPTER 3. SYSTEM MODEL AND SIMULATION SETUP

3.1 Introduction

This chapter presents the system model and simulation methodology used to evaluate the performance of OFDM and SC-FDMA transmission systems. Both systems were implemented in MATLAB under identical transmission conditions to ensure a fair comparison.

The simulations focus on two important performance metrics:

- Bit Error Rate (BER)
- Peak-to-Average Power Ratio (PAPR)

A multipath Rayleigh fading channel with additive white Gaussian noise (AWGN) was considered in the simulation. Zero Forcing (ZF) equalization was applied at the receiver to compensate for channel distortion.

3.2 Overall System Model

Both OFDM and SC-FDMA systems share highly similar transmission architectures. The major difference is the introduction of a DFT spreading operation in SC-FDMA before subcarrier mapping.

In OFDM, modulated symbols are directly mapped onto orthogonal subcarriers and transformed into the time domain using the Inverse Fast Fourier Transform (IFFT). In contrast, SC-FDMA first applies a DFT to the modulated symbols, which spreads the information symbols across the allocated subcarriers before OFDM modulation.

The OFDM transmitter structure can be summarized as:

Bits $\rightarrow$ 16-QAM $\rightarrow$ Subcarrier Mapping $\rightarrow$ IFFT $\rightarrow$ Cyclic Prefix $\rightarrow$ Channel

Meanwhile, the SC-FDMA transmitter introduces an additional DFT precoding stage:

Bits $\rightarrow$ 16-QAM $\rightarrow$ DFT $\rightarrow$ Subcarrier Mapping $\rightarrow$ IFFT $\rightarrow$ Cyclic Prefix $\rightarrow$ Channel

At the receiver side, both systems perform cyclic prefix removal and FFT processing to recover the transmitted subcarriers. Frequency-domain equalization is then applied to compensate for multipath fading effects.

For SC-FDMA, an additional inverse DFT (IDFT) operation is required after subcarrier demapping to recover the original symbol sequence.

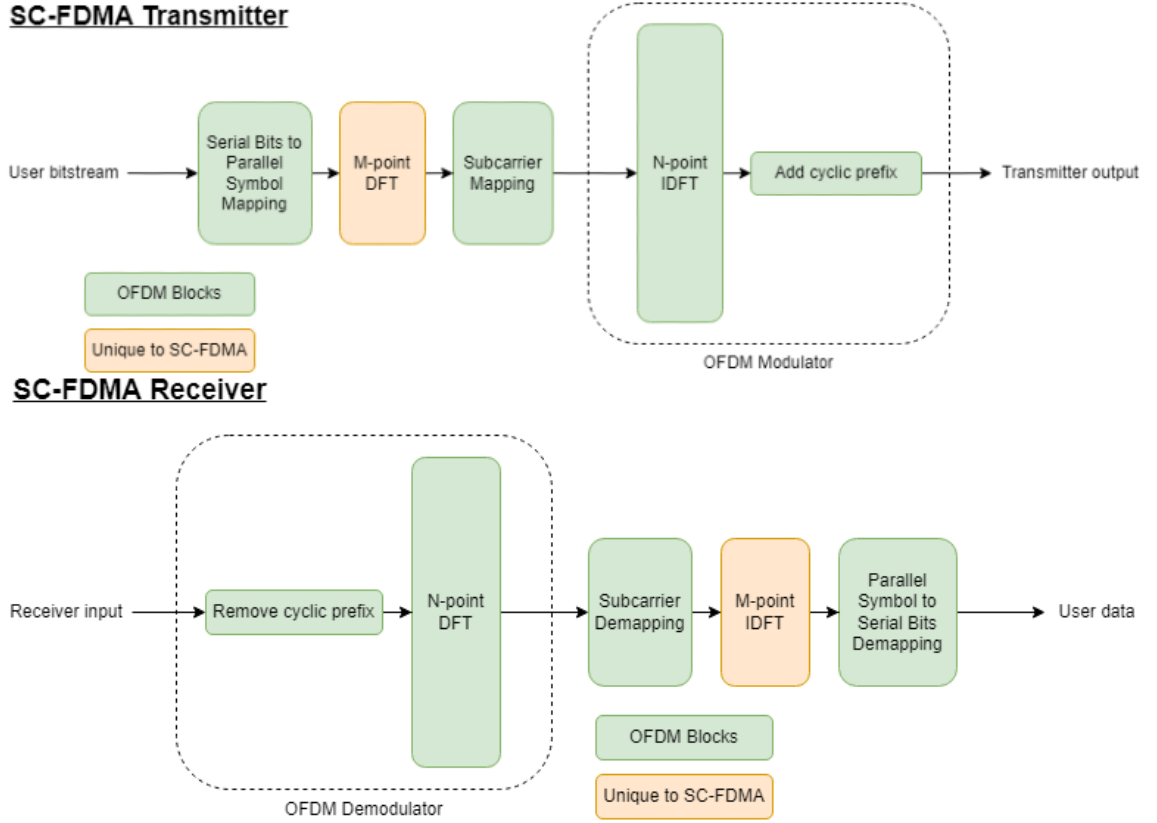


Figure 3.1 SC-FDMA transceiver structure based on OFDM modulation

In Figure 3.1, the green processing blocks correspond to conventional OFDM operations, while the orange blocks represent the additional DFT spreading and despreading operations unique to SC-FDMA.

The additional DFT precoding stage gives SC-FDMA its single-carrier-like characteristics and significantly reduces the Peak-to-Average Power Ratio (PAPR) compared to conventional OFDM systems.

3.3 Simulation Parameters

The simulation parameters used for both OFDM and SC-FDMA systems are listed in Table 3.1.

Table 3.1 Simulation Parameters

Parameter	Value
FFT size (N_{FFT})	64
Number of active subcarriers	16
Modulation scheme	16-QAM
Cyclic Prefix length	16
Channel type	Multipath Rayleigh fading
Equalization method	Zero Forcing (ZF)
SNR range	0 dB – 20 dB
Number of transmitted blocks	200000
Channel taps	3

A large number of transmitted blocks was selected to improve BER and CCDF statistical accuracy.

3.4 Transmitted Signal Generation

Random binary data were generated and grouped into symbols using 16-QAM modulation.

The number of transmitted bits is calculated as:

$$N_b = N_{blk} \times N_{used} \times \log_2(M) \quad (3.1)$$

where:

- N_{blk} is the number of transmitted blocks,
- N_{used} is the number of active subcarriers,
- M is the modulation order.

For OFDM, the modulated symbols were directly mapped onto active subcarriers before IFFT processing.

For SC-FDMA, the symbols first passed through a DFT spreading stage:

$$X[m] = \sum_{n=0}^{M-1} d[n] e^{-j \frac{2\pi mn}{M}} \quad (3.2)$$

The resulting spread symbols were then mapped onto OFDM subcarriers and transformed into the time domain using the IFFT.

3.5 Cyclic Prefix Insertion

To mitigate inter-symbol interference caused by multipath propagation, a cyclic prefix was inserted before transmission.

The cyclic prefix consists of copying the last portion of the OFDM symbol and appending it to the beginning:

$$x_{CP}[n] = [x(N - N_{CP}), \dots, x(N - 1), x(0), \dots, x(N - 1)] \quad (3.3)$$

where:

- N is the FFT size,
- N_{CP} is the cyclic prefix length.

The CP length was selected to exceed the channel delay spread in order to preserve subcarrier orthogonality.

3.6 Multipath Rayleigh Fading Channel

A fixed three-tap Rayleigh fading channel was used throughout the simulation.

The generated channel taps are:

$$h = [-0.6341 - 0.1139i, 0.4427 + 0.5667i, -0.2504 - 0.0712i] \quad (3.4)$$

The total channel power was normalized:

$$\sum_{l=0}^{L-1} |h_l|^2 = 1 \quad (3.5)$$

The received signal is expressed as:

$$r[n] = x[n] * h[n] + w[n] \quad (3.6)$$

where:

- $x[n]$ is the transmitted signal,
- $h[n]$ is the channel impulse response,
- $w[n]$ is AWGN noise.

The Signal-to-Noise Ratio (SNR) was varied from 0 dB to 20 dB to evaluate system performance under different noise conditions.

3.7 Frequency-Domain Equalization

After cyclic prefix removal and FFT processing, frequency-domain equalization was applied.

The received subcarrier signal is:

$$Y[k] = H[k]X[k] + N[k] \quad (3.7)$$

Zero Forcing equalization estimates the transmitted symbol using:

$$\hat{X}[k] = \frac{Y[k]}{H[k]} \quad (3.8)$$

This operation compensates for channel distortion by dividing the received signal by the channel frequency response.

For SC-FDMA, an additional IDFT despreading operation was applied after equalization:

$$d[n] = \frac{1}{M} \sum_{m=0}^{M-1} X[m] e^{j \frac{2\pi mn}{M}} \quad (3.9)$$

3.8 BER Evaluation Method

Bit Error Rate (BER) was computed by comparing the transmitted bits with the detected bits at the receiver.

$$\text{BER} = \frac{N_e}{N_t} \quad (3.10)$$

where:

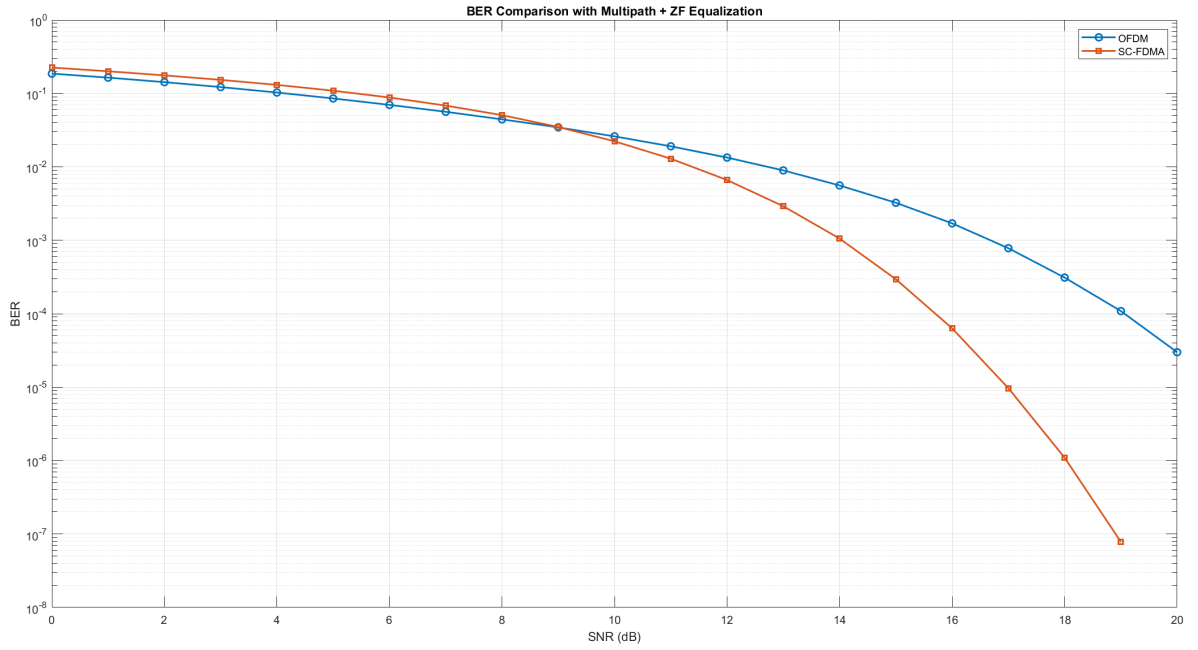
- N_e is the number of erroneous bits,
- N_t is the total number of transmitted bits.

The BER performance was evaluated for both OFDM and SC-FDMA systems over the entire SNR range.

Table 3.2 summarizes the obtained BER results.

Table 3.2 BER Performance Comparison

SNR (dB)	OFDM BER	SC-FDMA BER
0	1.859452×10^{-1}	2.249893×10^{-1}
5	8.538000×10^{-2}	1.088427×10^{-1}
10	2.599531×10^{-2}	2.228805×10^{-2}
15	3.242734×10^{-3}	2.948438×10^{-4}
20	2.992188×10^{-5}	0

**Figure 3.2 BER performance comparison between OFDM and SC-FDMA**

3.9 PAPR and CCDF Evaluation

PAPR measurements were performed for both OFDM and SC-FDMA transmitted signals.

The PAPR is defined as:

$$\text{PAPR} = \frac{\max |x[n]|^2}{E\{|x[n]|^2\}} \quad (3.11)$$

The Complementary Cumulative Distribution Function (CCDF) was used to evaluate the probability that the PAPR exceeds a specified threshold:

$$\text{CCDF}(P_0) = \Pr(\text{PAPR} > P_0) \quad (3.12)$$

The measured average PAPR values are:

Table 3.3 Average PAPR Comparison

System	Average PAPR (dB)
OFDM	6.0139
SC-FDMA	4.8222

The results clearly show that SC-FDMA achieves significantly lower PAPR than OFDM.

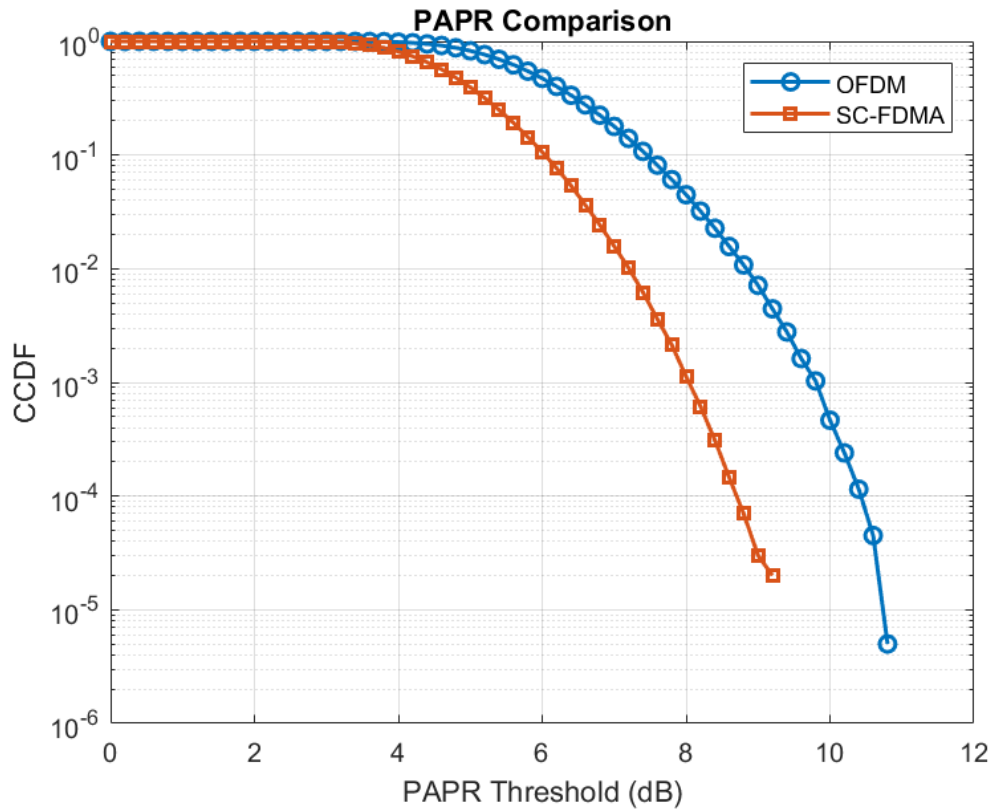


Figure 3.3 CCDF comparison of OFDM and SC-FDMA PAPR

3.10 Computational Considerations

A total of 200000 transmitted blocks were simulated for each SNR point. The simulation therefore required a large computational workload to obtain statistically reliable BER and CCDF results.

Progress indicators were implemented during execution to monitor the simulation status:

$$\text{SNR} = 10 \text{ dB} \mid \text{Processing block } 100000 / 200000$$

The use of a fixed channel realization ensured fair comparison between OFDM and SC-FDMA under identical fading conditions.

3.11 Chapter Summary

This chapter presented the MATLAB simulation framework used to compare OFDM and SC-FDMA systems under multipath Rayleigh fading channels.

The transmitter structure, cyclic prefix operation, channel model, frequency-domain equalization, BER analysis, and PAPR evaluation methods were described in detail.

The next chapter presents a detailed discussion and interpretation of the simulation results.

CHAPTER 4. CONCLUSION

4.1 Summary of the Study

This report presented a comparative study between Orthogonal Frequency Division Multiplexing (OFDM) and Single-Carrier Frequency Division Multiple Access (SC-FDMA), with emphasis on their Bit Error Rate (BER) performance and Peak-to-Average Power Ratio (PAPR) characteristics.

Both transmission schemes were implemented in MATLAB under identical simulation conditions, including:

- 16-QAM modulation
- FFT size of 64
- 16 active subcarriers
- Cyclic Prefix length of 16
- Multipath Rayleigh fading channel
- Zero-Forcing frequency-domain equalization

A large-scale Monte Carlo simulation consisting of 200,000 transmitted blocks was performed to ensure statistically reliable BER and PAPR measurements.

4.2 Discussion of BER Performance

Simulation results showed that both OFDM and SC-FDMA achieved reliable communication performance over the multipath fading channel.

At low SNR values, OFDM demonstrated slightly better BER performance than SC-FDMA. However, as the SNR increased, SC-FDMA gradually outperformed OFDM and achieved lower BER at medium and high SNR regions.

This behavior can be explained by the DFT spreading operation used in SC-FDMA. Since transmitted symbols are distributed across multiple subcarriers, the system becomes more robust against severe fading on individual subcarriers.

For example:

- At 0 dB SNR:

$$\text{BER}_{\text{OFDM}} = 1.859452 \times 10^{-1}$$

$$\text{BER}_{\text{SC-FDMA}} = 2.249893 \times 10^{-1}$$

- At 20 dB SNR:

$$\text{BER}_{\text{OFDM}} = 2.992188 \times 10^{-5}$$

$$\text{BER}_{\text{SC-FDMA}} = 0$$

The BER curves also showed the expected monotonic improvement as SNR increased, indicating that the system implementation and equalization process operated correctly.

4.3 Discussion of PAPR Performance

The PAPR analysis demonstrated one of the most important advantages of SC-FDMA over OFDM.

The average measured PAPR values were:

$$\text{PAPR}_{\text{OFDM}} = 6.0139 \text{ dB} \quad (4.1)$$

$$\text{PAPR}_{\text{SC-FDMA}} = 4.8222 \text{ dB} \quad (4.2)$$

Therefore, SC-FDMA achieved approximately:

$$\Delta\text{PAPR} = 6.0139 - 4.8222 = 1.19 \text{ dB} \quad (4.3)$$

of average PAPR reduction compared to OFDM.

The CCDF curves further confirmed that OFDM experiences significantly higher probabilities of large instantaneous power peaks.

For example, at a PAPR threshold near 8 dB:

- OFDM still exhibited noticeable probability of occurrence
- SC-FDMA probability had already approached zero

This lower PAPR property is highly important in practical wireless transmitters because it improves power amplifier efficiency and reduces nonlinear distortion.

4.4 Engineering Implications

The results explain why OFDM and SC-FDMA are used differently in practical cellular systems.

OFDM is highly suitable for downlink transmission because base stations have:

- Large power budgets
- More sophisticated hardware
- Less stringent amplifier efficiency constraints

In contrast, uplink mobile devices operate under limited battery capacity. Therefore, reducing PAPR becomes critically important.

SC-FDMA preserves many OFDM advantages such as:

- Frequency-domain equalization
- Robustness against multipath fading
- Flexible subcarrier allocation

while simultaneously improving transmitter power efficiency through its single-carrier-like waveform characteristics.

This is the primary reason why SC-FDMA was selected for LTE uplink transmission systems.

4.5 Limitations of the Study

Although the implemented simulation successfully demonstrates the fundamental differences between OFDM and SC-FDMA, several simplifying assumptions were made:

- Perfect synchronization was assumed
- Perfect channel estimation was assumed
- Only localized SC-FDMA mapping was considered
- A fixed modulation order (16-QAM) was used
- Nonlinear power amplifier effects were not modeled

In practical communication systems, these factors may further influence system performance.

4.6 Future Work

Several possible extensions can be explored in future studies:

- Comparison between Localized and Interleaved SC-FDMA
- Investigation of higher-order modulation schemes
- Analysis under imperfect channel estimation
- Evaluation with nonlinear power amplifier models
- Comparison under realistic LTE or 5G numerologies
- Study of spectral efficiency and computational complexity

4.7 Final Remarks

In conclusion, this study demonstrated the fundamental tradeoff between OFDM and SC-FDMA.

OFDM provides excellent spectral efficiency and implementation simplicity, but suffers from high PAPR due to the superposition of many independently modulated subcarriers.

SC-FDMA mitigates this problem through DFT spreading, producing lower PAPR and improved transmitter efficiency while retaining most OFDM advantages.

The simulation results verified that SC-FDMA achieves significantly lower PAPR while maintaining competitive BER performance under multipath fading conditions. These characteristics make SC-FDMA highly suitable for uplink wireless communication systems where power efficiency is critical.

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APPENDIX

Source Code Repository

The complete source code used in this report is publicly available on GitHub. The repository contains:

- MATLAB simulation source code
- BER and PAPR evaluation scripts
- LaTeX source files for the report
- Generated simulation figures and plots

The repository can be accessed at:

<https://github.com/minh18ndh/ofdm-vs-scfdma>

The repository was created to improve reproducibility and allow future extensions of the simulation framework, including:

- Localized and Interleaved SC-FDMA
- Higher-order modulation schemes
- Channel estimation techniques
- Nonlinear power amplifier modeling
- LTE and 5G NR parameter configurations